Proposed Solution

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	4 Marks

Project Title

Personalized Travel Itinerary Planner

Solution Overview

An AI-powered, full-stack web application that enables travelers to generate personalized, budget-aware, day-by-day travel itineraries using Generative AI (Google Gemini / Groq LLMs). Users provide basic trip parameters and receive a complete, customizable travel plan in seconds.

Core Solution Components

1. AI Itinerary Engine

- **Primary Model**: Google Gemini 2.5 Flash — chosen for speed, structured JSON output, and cost efficiency
- **Fallback Model**: Groq LLama 3.3 70B Versatile — automatic failover when Gemini is unavailable
- **Prompt Engineering**: Carefully crafted prompts that enforce:
- Exact number of days matching the trip duration
- Real places at the specified destination
- Budget-aware cost estimation per activity
- Travel style compliance (3-7 activities/day based on style)
- Structured JSON output matching Pydantic schemas
- **Single Day Regeneration**: Separate prompts for regenerating individual days that avoid repeating activities from other days

2. RESTful API Backend (FastAPI)

- **Endpoints**: Health check, user profile, trip CRUD, trip generation, day regeneration, activity CRUD, activity reordering
- **Authentication**: JWT Bearer token verification using Supabase JWKS (supports key rotation)
- **Authorization**: Per-request Supabase client scoped to the authenticated user's token, enforcing RLS policies
- **Error Handling**: Centralized exception handlers for validation errors, auth failures, and 404s
- **Request Logging**: Unique request IDs with timing information for every HTTP request

3. PostgreSQL Database (Supabase)

- **Schema**: 6 tables — profiles, trips, itinerary_days, activities, trip_members, generation_logs
- **Row-Level Security**: 14 RLS policies ensuring data isolation between users
- **Helper Functions**: is_trip_owner(), has_trip_access(), can_edit_trip(), day_trip_id() — SECURITY DEFINER functions for safe policy evaluation
- **Triggers**: Auto-create profile on signup, auto-update timestamps on trip modification

4. React Frontend

- **Framework**: React 19 + TypeScript + Vite for modern, fast development
- **UI Library**: shadcn/ui components (Button, Card, Input, Label, Dialog) with TailwindCSS v4
- **State Management**: React Query (TanStack) for server state with automatic caching and invalidation
- **Forms**: React Hook Form + Zod for type-safe form handling and validation
- **Drag & Drop**: dnd-kit for accessible, performant activity reordering
- **Routing**: React Router DOM with protected routes and auth guards

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User Workflow

1. **Sign Up / Log In** → Supabase Auth handles registration and session management
2. **Create Trip** → Fill in destination, dates, budget, style, interests
3. **Generate Itinerary** → AI creates complete day-by-day plan with activities and costs
4. **Review & Customize** → Browse days, edit/delete activities, drag to reorder
5. **Regenerate Days** → Replace any day's plan with fresh AI suggestions
6. **Dashboard** → View all trips, track budgets, manage trip collection

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Deployment Architecture

Component	Platform	URL
Frontend	Vercel	https://personalized-travel-itenerary-planner-ob42m5s4v-tanay-e4be.vercel.app/
Backend API	Render	https://personalized-travel-itenerary-planner.onrender.com
API Docs	Render (Swagger)	https://personalized-travel-itenerary-planner.onrender.com/docs
Database	Supabase	Project ID: vskaghlmvjqambnrshhy
Source Code	GitHub	https://github.com/Tanay-Ds/Personalized-Travel-Itenerary-Planner